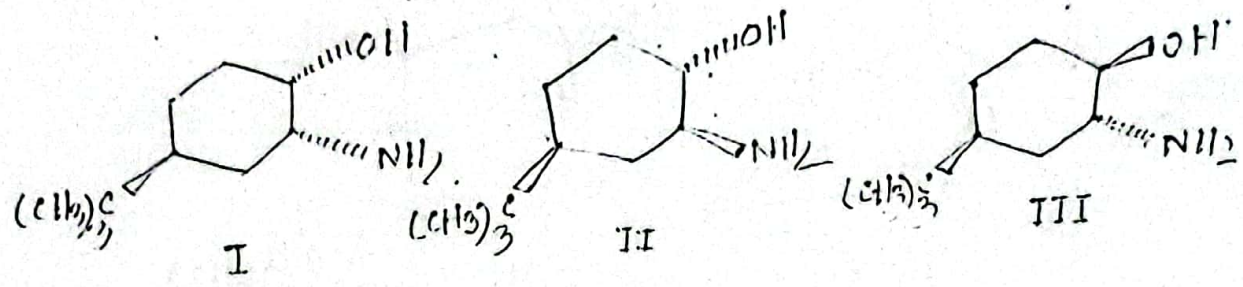
Leela ②

Configuration and conformation of cyclic compounds

*(i) Draw the chair form of the following compounds --

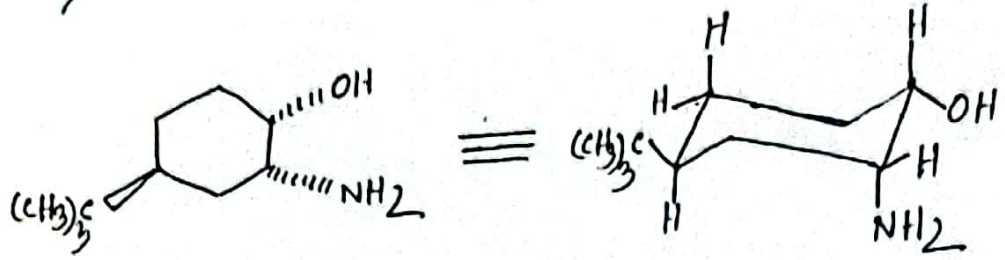


(ii) which one of these compounds is least stable? Give explanation.

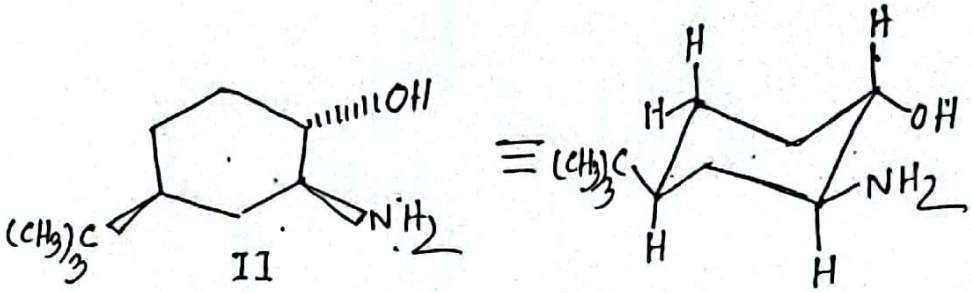
(iii) Are all of these compounds formed the same product by the reaction with nitrous acid? Give chemical reactions and explanation in support of your answer.

(iii) Is there any function of the tert-butyl group that present in each of the molecules?

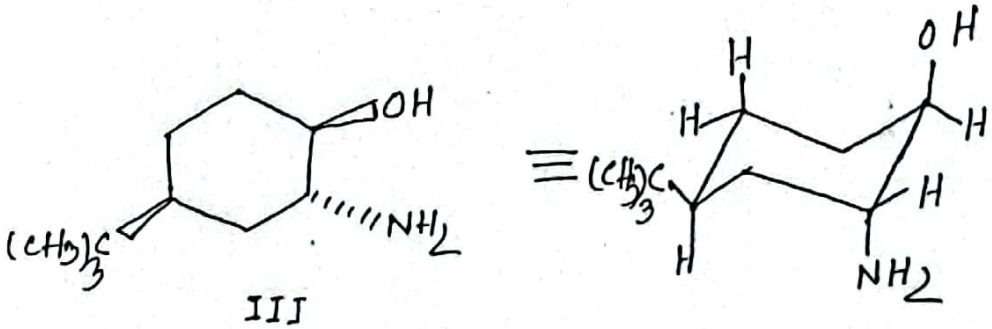
07



I

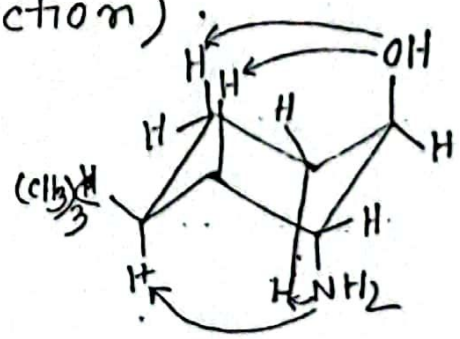


II

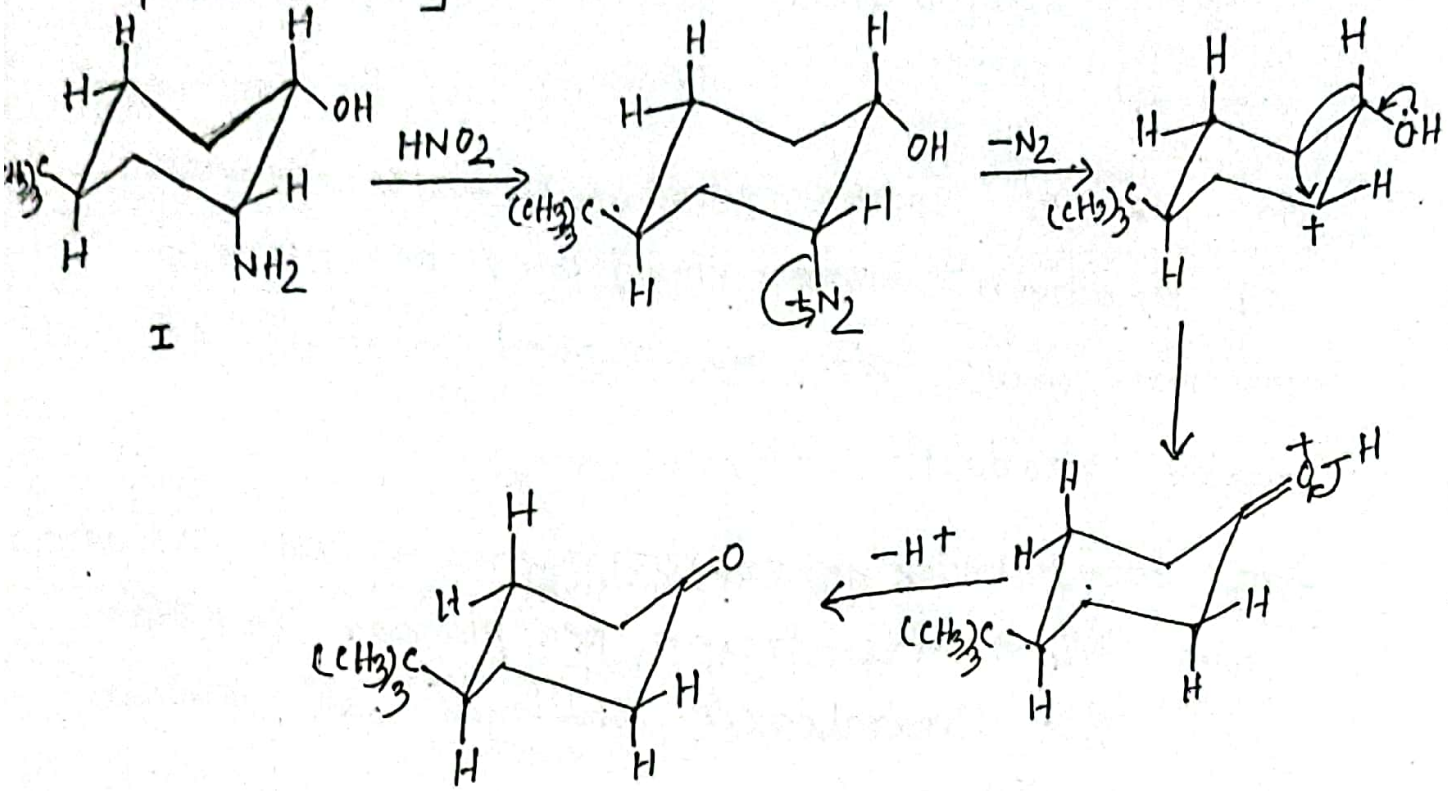


III

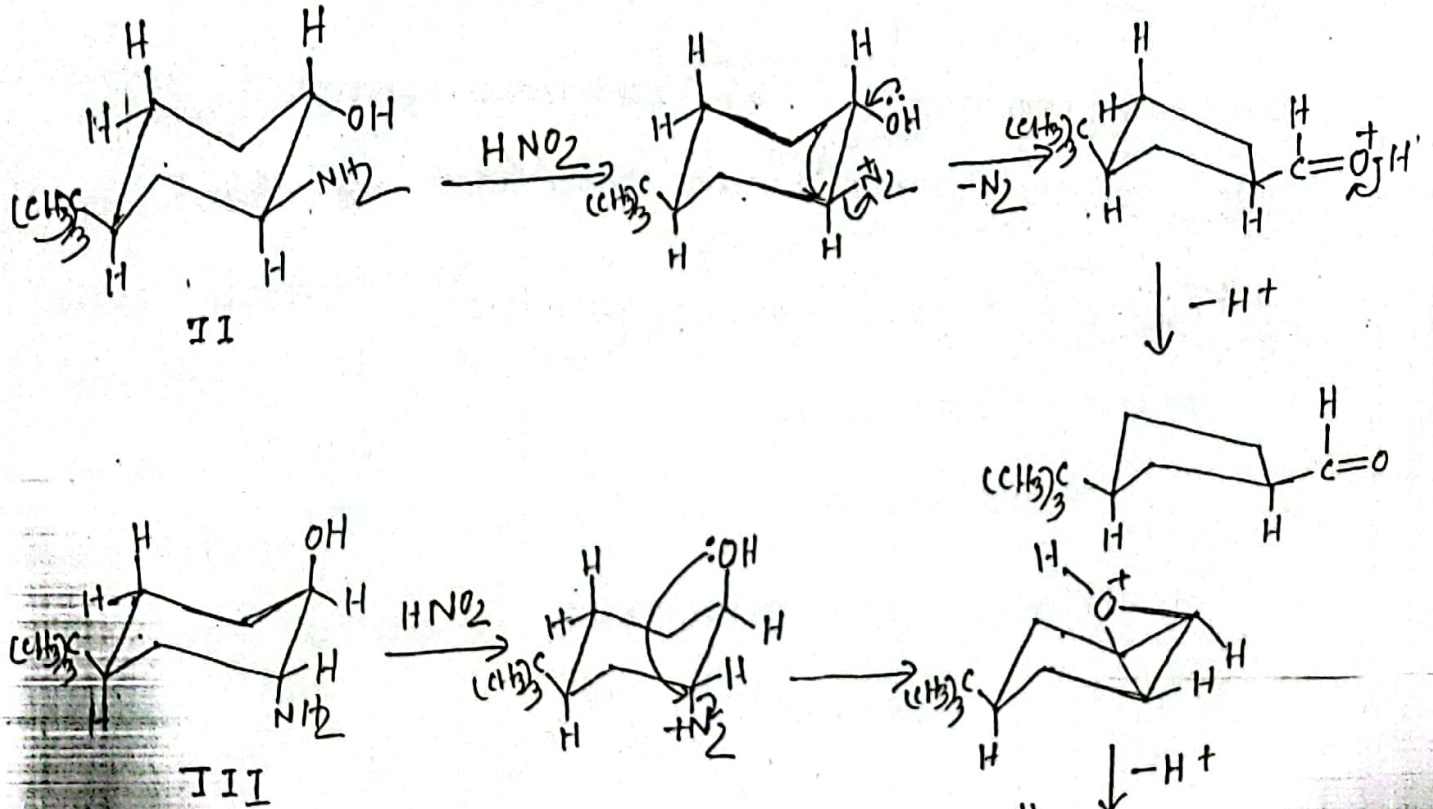
(i) Here compound III is least stable because it faces strongest 1,3-diaxial interaction (two OH–H and two NH_2–H diaxial interaction).



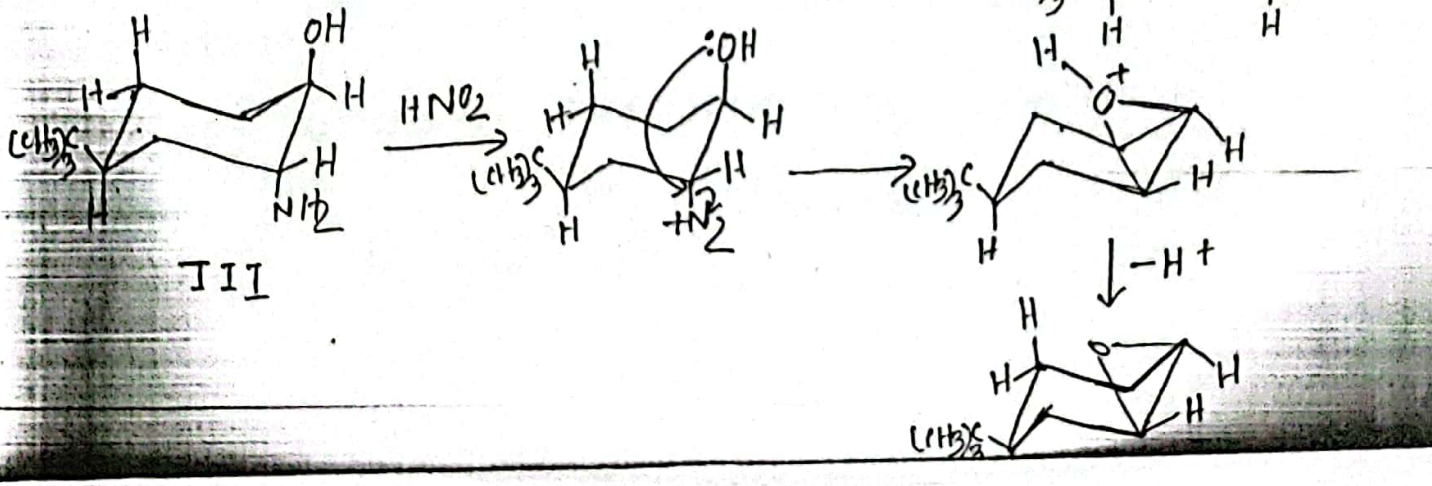
(ii) All of these compounds formed different product by reaction with nitrous acid.



I



II



III

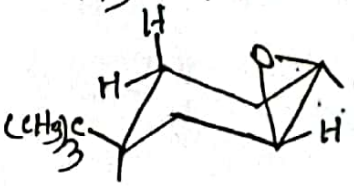
Explanation:

Here rearrangement occurs from its trans periplanar position ~~gives~~ for better overlap.

In case of compound-1, rearrangement of H-atom from its trans-periplanar position gives six membered ketone as major product.

For compound-2, rearrangement of carbon atom from its trans periplanar position gives five membered aldehyde as only product.

For compound-3, by rearrangement from its trans periplanar position gives



as major product.

iii) Yes. Here tert-butyl group act as an anchor. It lock the molecule in equitorial position. Thus, prevent the ring flipping.

* Explain with example fundamental basis of curtin-Hammett principle with regard to the proportioner of more than one product in a reaction.

The curtin-Hammett principle:

"It states that, for a reaction that has a pair of reactive intermediates or reactants that interconverted rapidly, each going irreversibly to a different product, the product ~~to~~ ratio will depend only on the difference in free energy ~~to each~~ product of the transition state going to each product and not on the equilibrium comtant between the intermediates."

Let us consider the following reaction-

$$P_A \xleftarrow[\text{slow}]{k_1} A \underset{\text{fast}}{\overset{k_A}{\rightleftharpoons}} B \xrightarrow[\text{slow}]{k_2} P_B \quad \text{.....(i)} \quad k_1, k_2 << k_A, k_B$$

Using rate eq^{n}(s) for eq^{n} (i) we get-

$$\frac{d[P_A]}{dt} = k_1[A]$$

$$\frac{d[P_B]}{dt} = k_2[B]$$

$$\therefore \frac{d[P_B]}{d[P_A]} = \frac{k_2[B]}{k_1[A]}$$

$$\Rightarrow d[P_B] = \frac{k_2[B]}{k_1[A]} \times d[P_A] \quad \ldots\ldots (ii)$$

since A and B are in equilibrium-

$$K_{eq} = \frac{[B]}{[A]}$$

Now eq^{n} (ii) becomes-

$$d[P_B] = \frac{k_2}{k_1} \cdot K_{eq}\, d[P_A]$$

By integrating, we get-

$$\int d[P_B] = \int \frac{k_2}{k_1} K_{eq}\, d[P_A]$$

$$\Rightarrow [P_B] = \frac{k_2}{k_1} \cdot K_{eq} \cdot [P_A]$$

$$\Rightarrow \frac{[P_B]}{[P_A]} = \frac{k_2}{k_1} K_{eq} \quad \ldots\ldots (iii)$$

From the relation between equilibrium constant and free energy we know-

$$\Delta G^{\circ} = -RT \ln K_{eq}$$

$$\Rightarrow K_{eq} = e^{-\Delta G^{\circ}/RT}$$

$$\therefore k_1 = e^{-\Delta G_1^{\neq}/RT}, \quad k_2 = e^{-\Delta G_2^{\neq}/RT}$$

Now equation (iii) becomes-

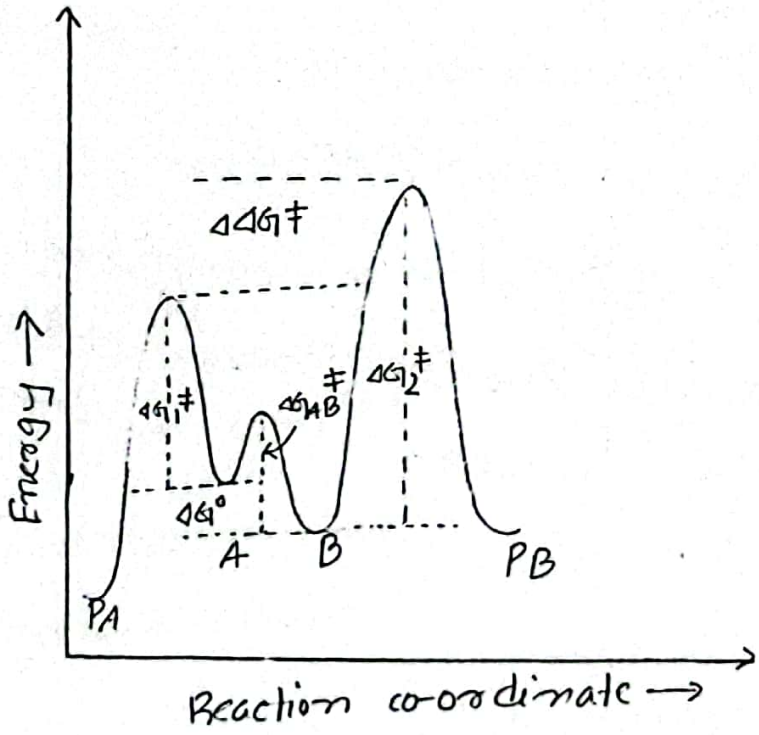
$$\frac{[P_B]}{[P_A]} = \frac{e^{-\Delta G_2^{\ddagger}/RT}}{e^{-\Delta G_1^{\ddagger}/RT}} \; e^{-\Delta G^{\circ}/RT}$$

$$= e^{-(\Delta G_2^{\ddagger} + \Delta G^{\circ} - \Delta G_1^{\ddagger})/RT}$$

$$= e^{-\Delta\Delta G^{\ddagger}/RT}$$

$$= e$$

where $\Delta\Delta G^{\ddagger} = \Delta G_2^{\ddagger} + \Delta G^{\circ} - \Delta G_1^{\ddagger}$



** Draw the chair form and the corresponding Newman projection formula of cyclohexane and cyclohexanone and give satisfactory answer the following questions-

i) Is there any eclipsed hydrogen atom in the cyclohexane?

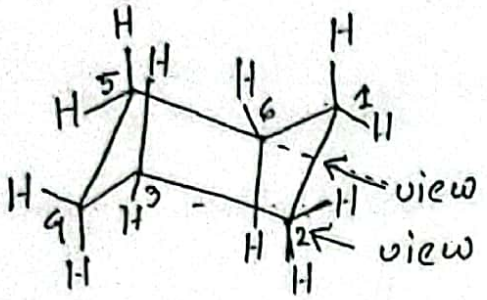
ii) Is there any eclipsed hydrogen atom in the cyclohexanone?

iii) Is the cyclohexanone molecule free from angle strain?

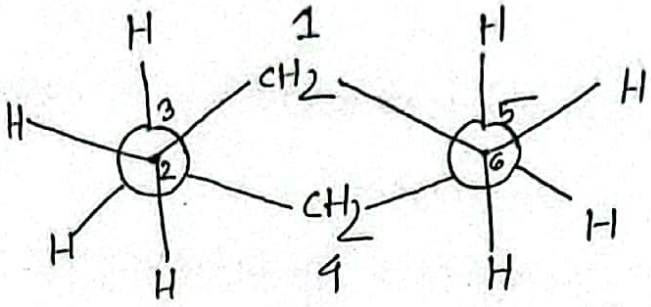
iv) Which one of the compounds is more stable? Give explanation in support of your answer.

a)

Chair form and corresponding Newman projection formula of cyclohexane:

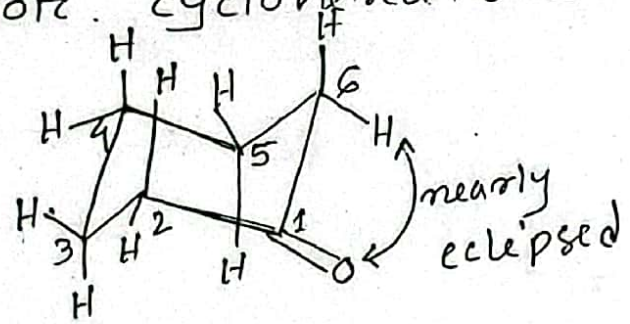


Chair conformation of cyclohexane

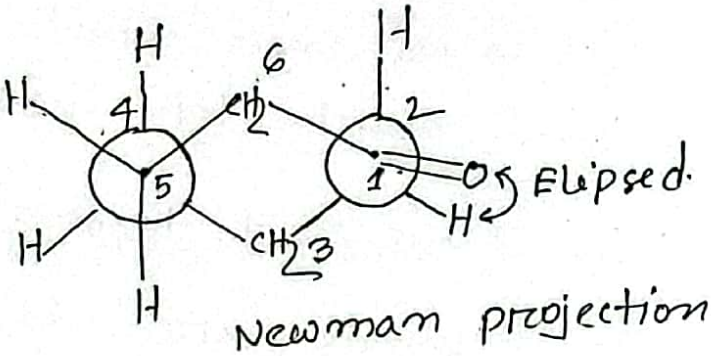


Newman projection of chair conformation (cyclohexane)

For cyclohexanone—



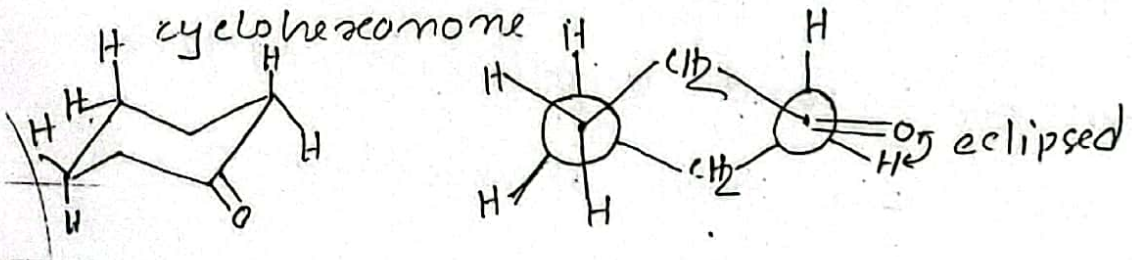
Chair conformation of cyclohexanone



Newman projection of cyclohexanone

i) There is no eclipsed hydrogen atom in the cyclohexane.

ii) There is eclipsed hydrogen atom in the cyclohexanone

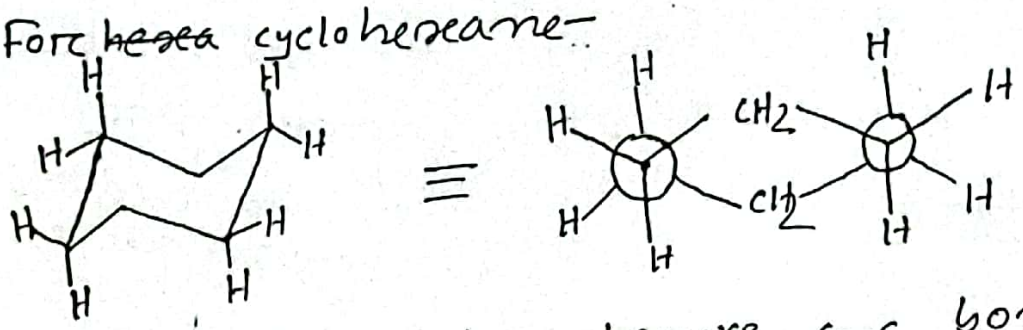


iii) No. The cyclohexanone molecule is not free from angle strain due to deviation from ideal bond angle 109.5°.

iv) cyclohexane is more stable than cyclohexa

Because-

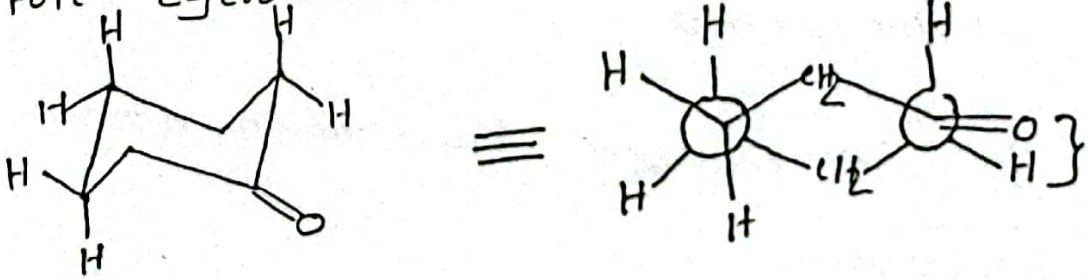
For chair cyclohexane-



→ No angle strain. because C–C bonds are nearly tetrahedral and bond angle is 109°

→ Minimal torsional strain because C–C bonds are staggered.

For cyclohexanone-



→ Angle strain is present due to deviation from ideal bond angle 109.5°

→ since C–C bond length in cyclohexanon is lower than that of cyclohexane, the

equitorial H atom is nearly placed at O atom. As a result torsional strain is observed between eclipsing H and O atom.

For this reason, cyclohexane is more stable than cyclohexanone.

* Draw the boat conformation and its corresponding Newman projection formula of cyclohexane and give answer of the following questions:

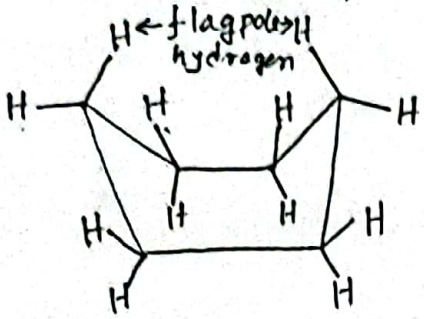
i) How many pairs of eclipsed hydrogen atoms does it have?

ii) How many flag pole hydrogen does it have?

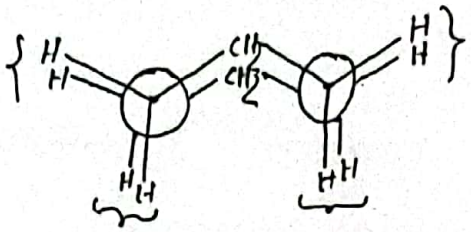
iii) what kind of strain occurs between the eclipsed hydrogen atoms?

iv) what kind of interaction occurs between flag-pole hydrogen atoms?

a) Boat conformation and corresponding Newman projection formula of cyclohexane.



Boat conformation



Newman projection of boat conformation

i) cyclohexane has four pairs of eclipsed hydrogen atoms.

ii) It has two flagpole hydrogen atoms.

iii) Torsional strain occurs between eclipsed hydrogen atoms.

iv) Flag pole interaction (Van der waals strain) occurs between flag pole hydrogen atoms.

* Q) After puckering the planar polyhexagone formula of cyclohexane there exists two non-planar structures which are free from angle strain.

i) what are the names of these two non-planar structures of cyclohexane?

ii) Draw these two non-planar structures of cyclohexane.

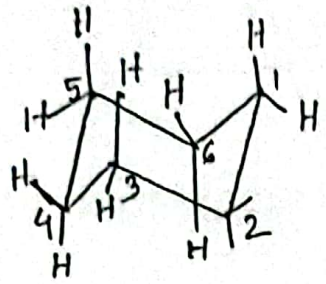
iii) which one of these two non planar structures of cyclohexane is more stable? Give reason in support of your answer.

iv) which one of these two non planar structures of cyclohexane is less stable? Give reason in support of your answer.

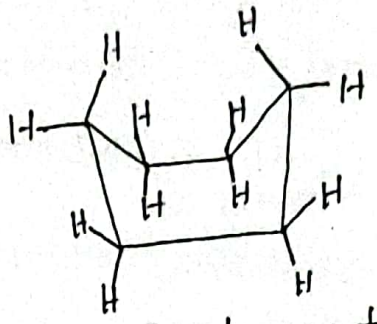
i) Names of two non planar structures of cyclohexane are-

Chair conformation and boat conformation.

ii)

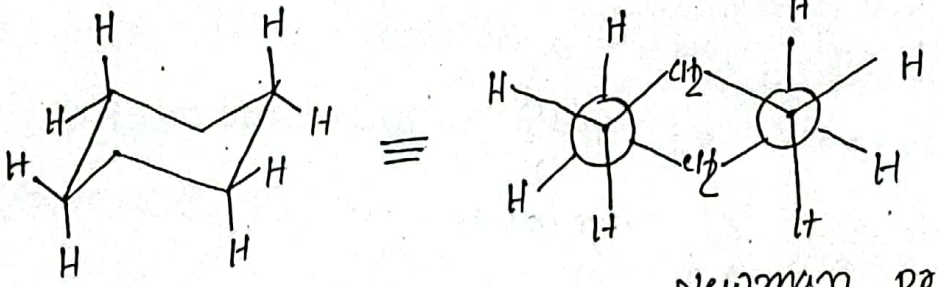


Chair conformation



Boat conformation

iii) Chair conformation is more stable than boat conformation.



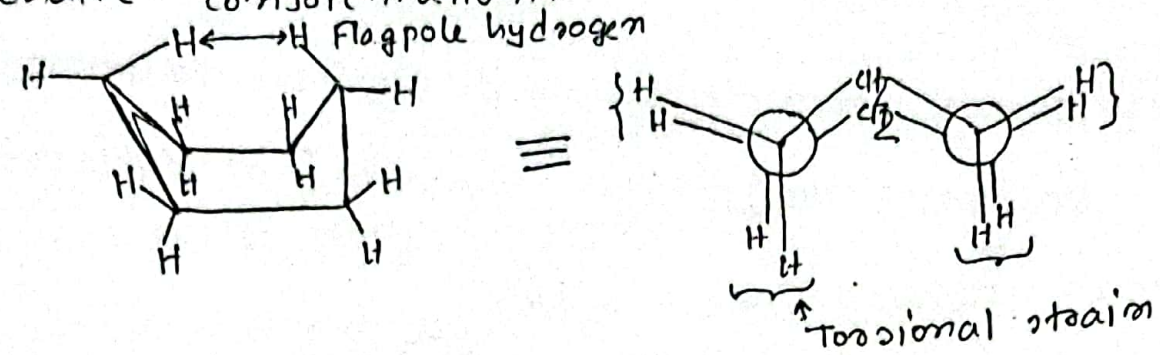
Newman projection

→ No angle strain because C-C bonds are nearly tetrahedral and bond angle is 109°

→ Minimal torsional strain because C-C bonds are staggered.

→ No ring strain (0 Kcal/mol)

iv) Boat conformation is less stable than chair conformation.



→ No angle strain but close contact between flagpole hydrogen causes van der waals strain in boat.

→ Huge torsional strain is present because of having 4 pairs of eclipsing H-atoms.

→ Have ring strain (7.0 kcal/mol)

* a) write down the name of the smallest member of cycloalkane.

b) Draw the planar structure and its corresponding Newman projection formula of the smallest cycloalkane and give answer of the following questions—

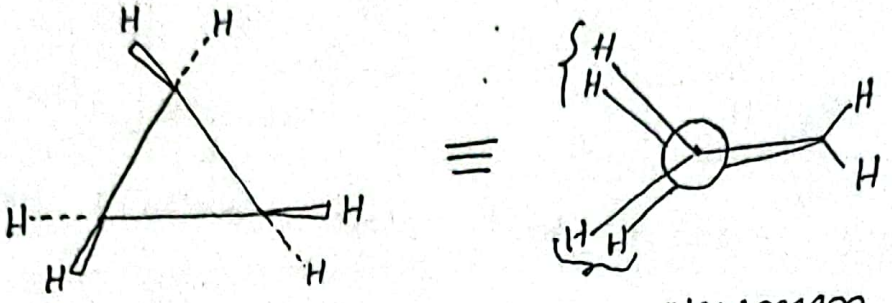
i) Is the cycloalkane free from angle strain?

ii) Is there any eclipsed CH bonds?

iii) What kind of strain present between the bonds of the cycloalkane?

a) smallest member of cycloalkane is cyclopropane.

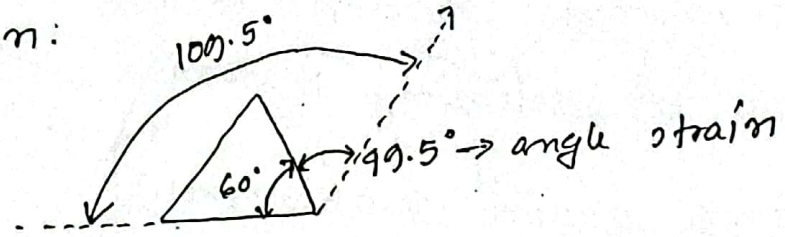
b) Planar structure and its corresponding Newman projection formula of cyclopropane



Planar structure of cyclopropane

Newman projection of cyclopropane

i) Cyclopropane is not free from angle strain:



ii) Yes. There is two pairs of eclipsing C-H bonds.

iii) Torsional strain is present between the eclipsing C-H bonds.

a)
* Draw the planar and non planar structures of cyclobutane and their corresponding Newman projection formula and give answer of the following questions-

i) Is there any angle strain in the planar and non planar structures of cyclobutane?

ii) which one of these two structures has higher angle strain?

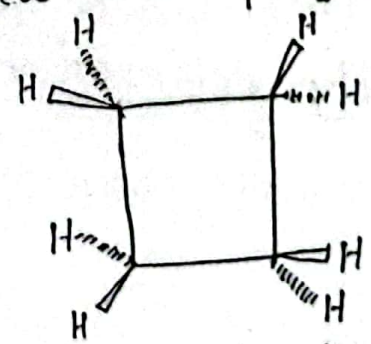
iii) which one of these two structures has strong torsional strain.

iv) which one of these two structures of cyclobutane is more stable?

Ans:

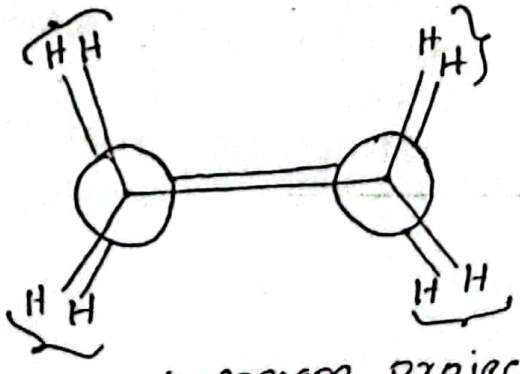
a)

Planar structure of cyclobutane and corresponding Newman projection formula-



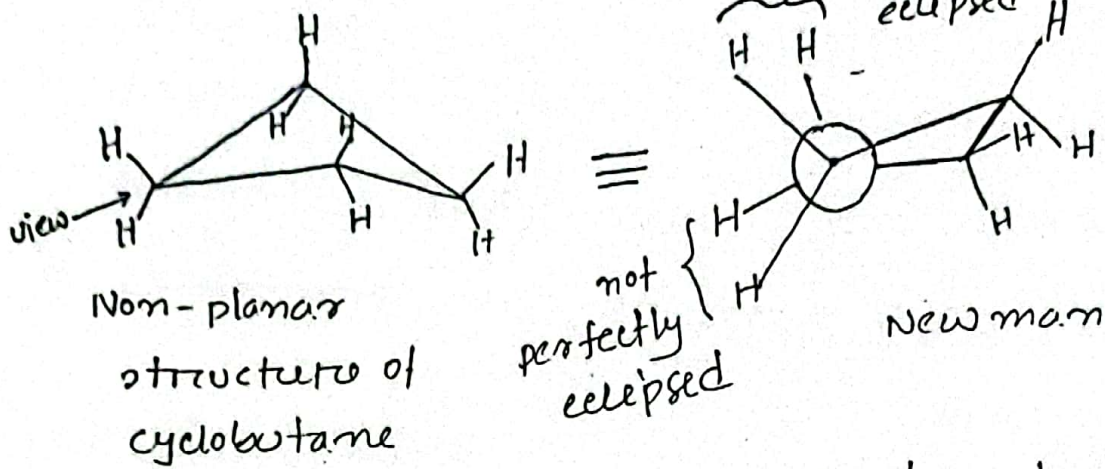
Planar structure of cyclobutane

≡



Newman projection

Non-planar structure of cyclobutane and corresponding Newman projection formula-



Non-planar structure of cyclobutane

Newman projection

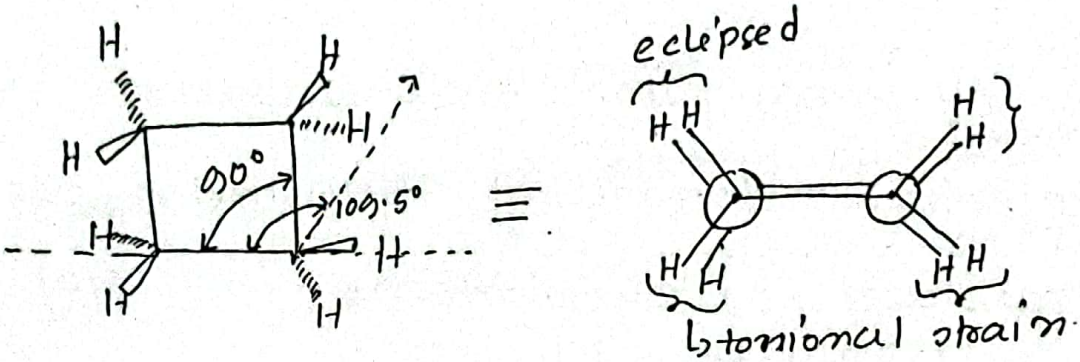
i) Yes. Angle strain present in both planar and non planar structure of cyclobutane due to deviation from ideal bond angle 109.5°

ii) Non planar structure of cyclobutane has higher angle strain than planar structure.

iii) Planar structure of cyclobutane has higher torsional strain (8 pairs of eclipsing H atoms than non planar structure (two pairs slightly eclipsed H atoms.

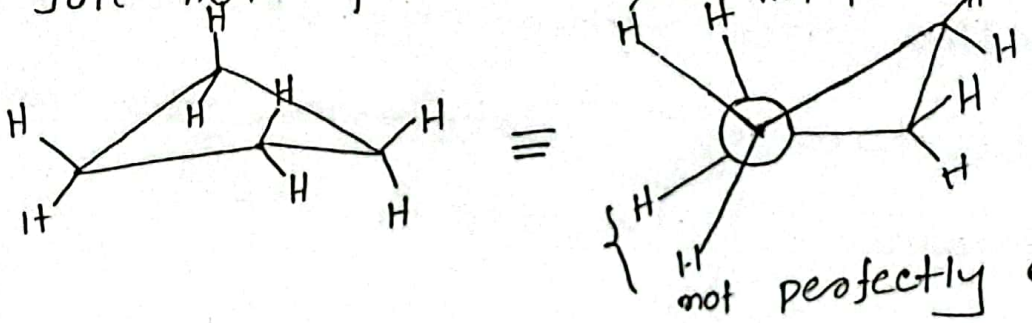
iv) Non planar structure of cyclobutane is more stable than planar structure.

For planar structure—



∴ it faces angle strain (19.5°) and huge torsional strain because of having four pairs of eclipsing H atoms.

But for non planar structure—



Here torsional strain is almost reduced by adopting a ~~set~~ slightly folded structure but this results in some additional angle strain.

v) Are the C–H bonds in the newman projection formula are perfectly eclipsed or not in non planar structure of cyclobutane?

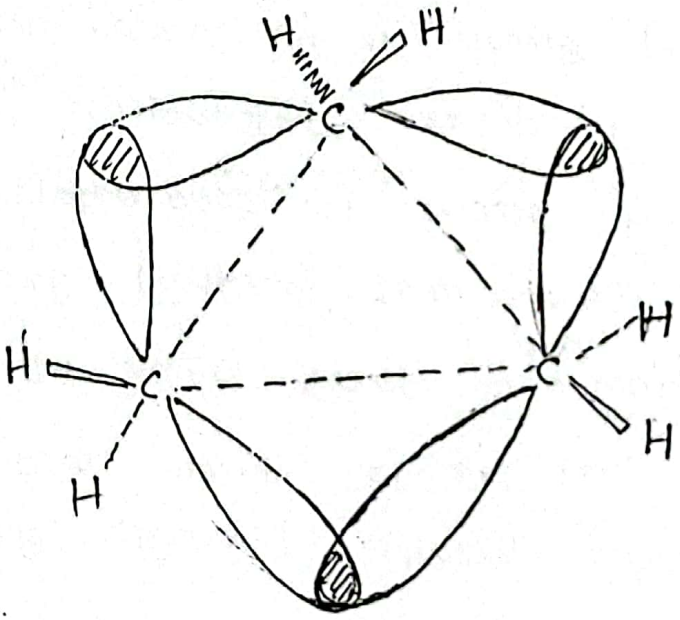
→ The CH bonds in the Newman projection formula for non-planar structure of cyclobutane is not perfectly eclipsed.

vi) As the angle strain increased or decrea after the puckering of the planar structure of cyclobutane.

→ After the puckering of the planar structure of cyclobutane, the angle strain is increased.

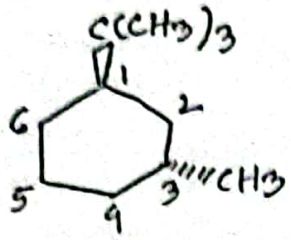
* cyclohexane has a puckered shape like a lounge chair rather than a flat shape. why.

** Although cyclopropane ~~has~~ faced huge angle strain in planar conformation but it exists in planar conformation. Because the bonding electrons lie principally outside the triangular internuclear lines resulting in bent bonds.

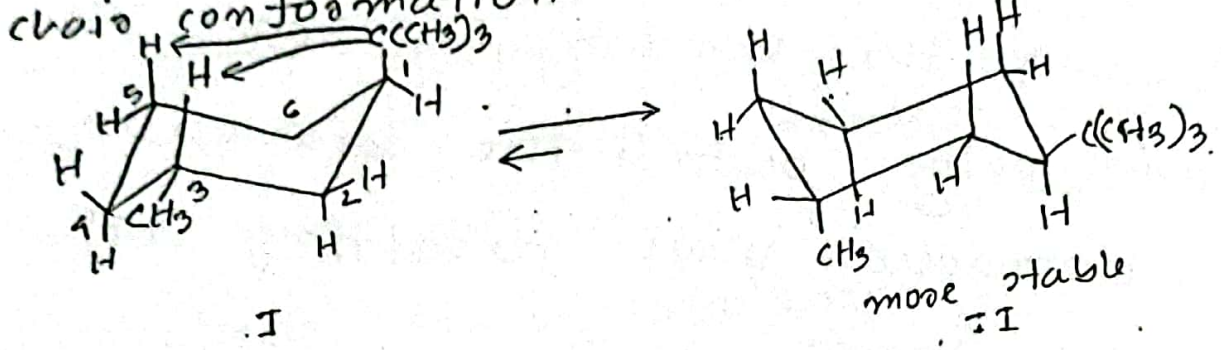


* Draw the chair conformations of the following compounds and explain which conformer in each case is more stable.

(i) trans-1-t-butyl-3-methylcyclohexane.



chair conformation-



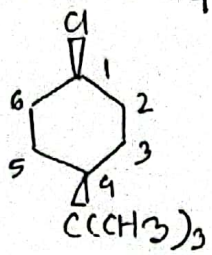
I

more stable

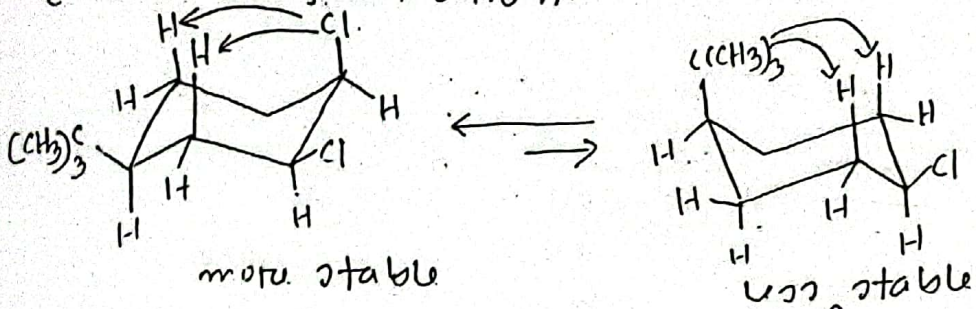
II

when t-butyl group is in axial position it faces more 1,3-diaxial interaction than when tertiary group in equatorial position. Although in structure II, methyl group is in axial position and it faces some 1,3-diaxial interaction but it is much less than t-butyl group because t-butyl group is larger than methyl group.

(ii) cis-1-chloro-4-t-butyl cyclohexane.



chair conformation-



more stable
A

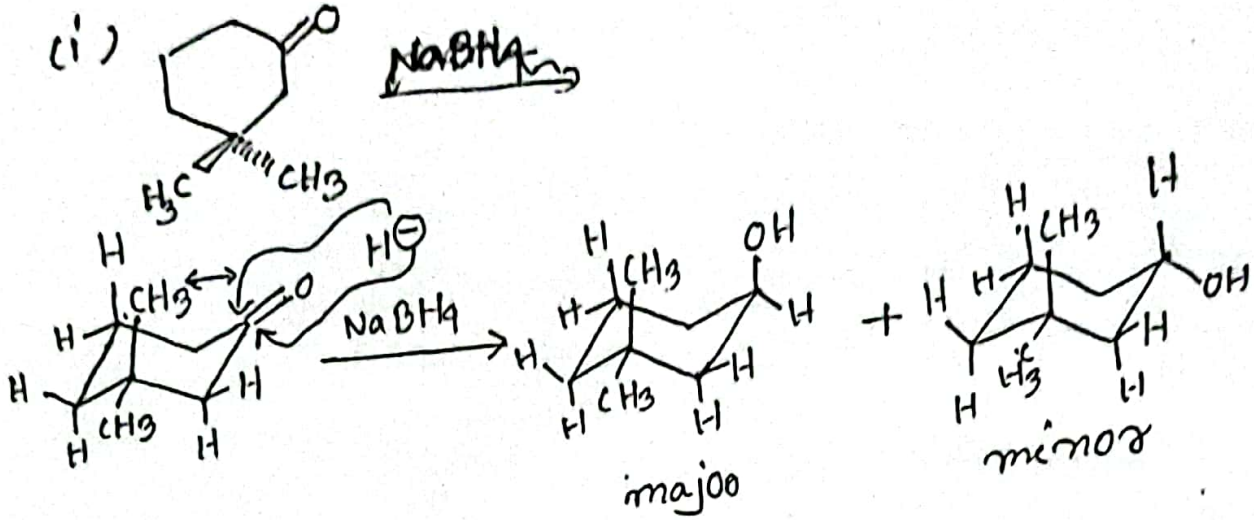
less stable
B

Here tertiary butyl group is bigger than a. As a result, when t-butyl group is in axial direction it faces more 1,3-diaxial interaction than when a group is in axial direction.

For this reason, A is more stable.

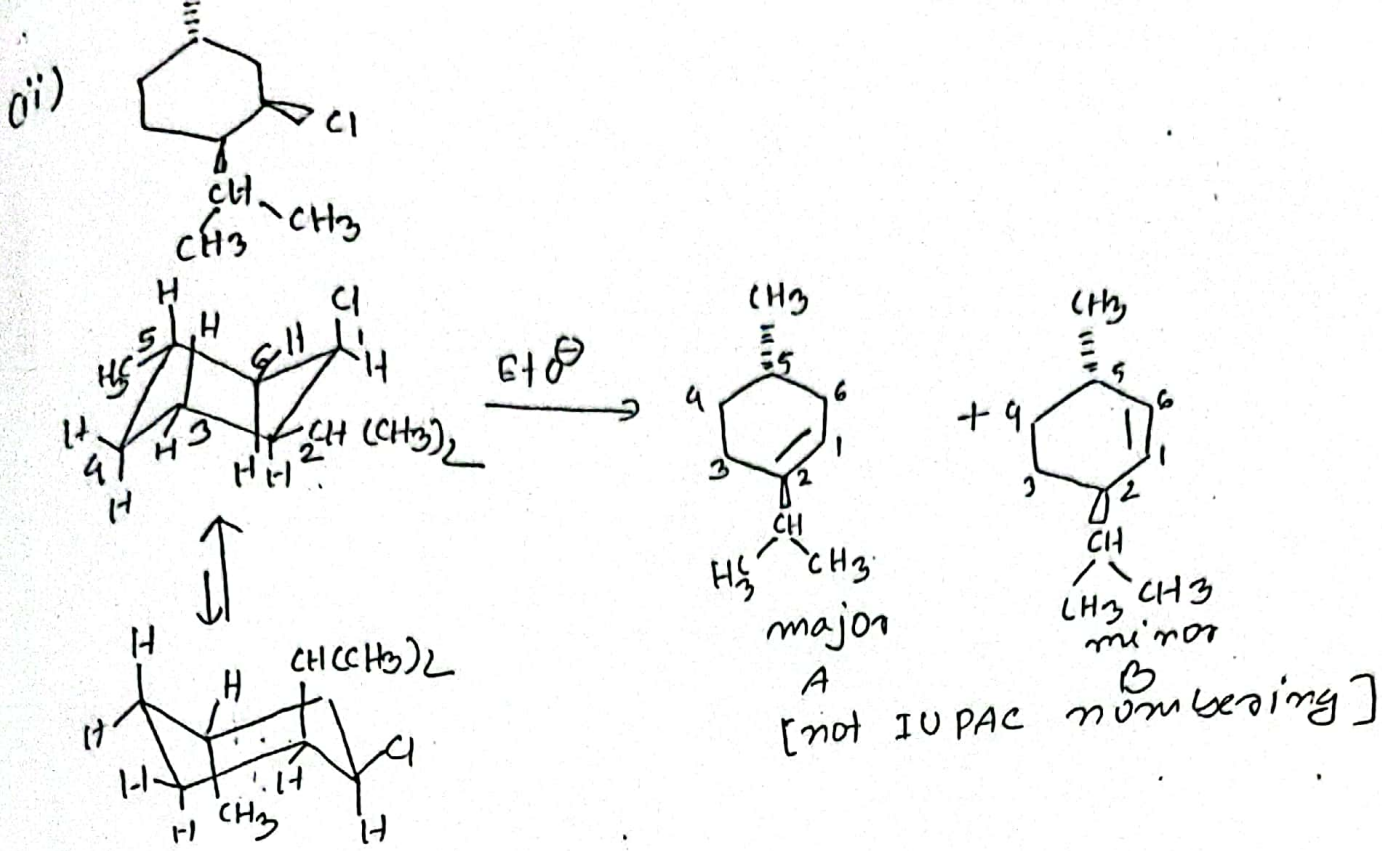
+ Predict the major product of the following reactions and give explanation in favour of your answer.

(i)



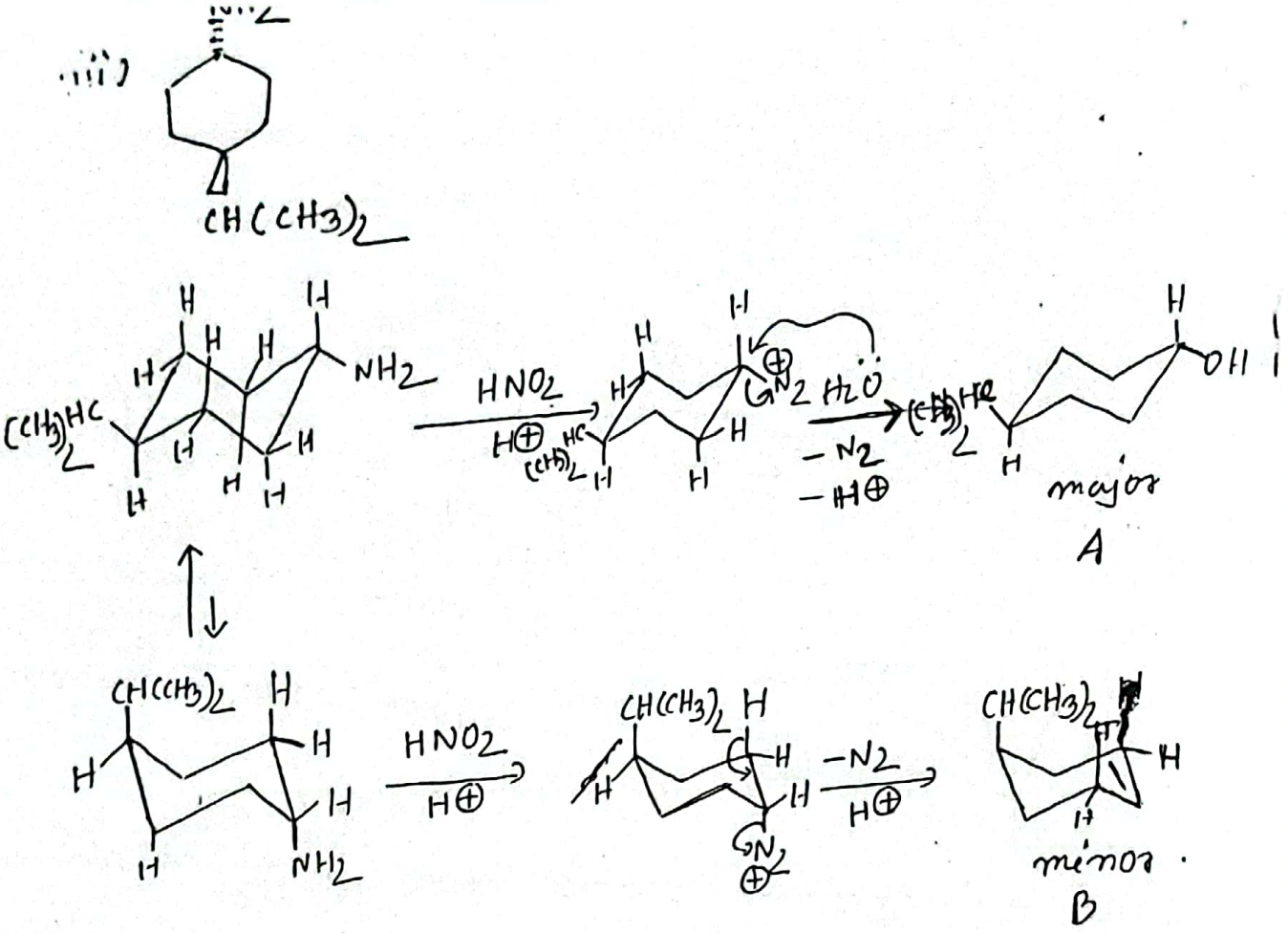
we know hydride ion is likely to attack from the less hindered site, when hydride ion attack in axial direction leading to equatorial alcohol it faces more steric ~~hindrance~~ strain than when hydride ion attack in equatorial direction leading to axial alcohol. Although when hydride ion attack in equatorial direction it faces torsional strain but here steric strain predominate torsional strain.

(ii)



[not IUPAC numbering]

Here elimination occurs from trans periplanar position. But for Cl trans periplanar position is 2 and 6. So elimination can occurs from position 2 and 6. But elimination predominantly occurs from position 2, because from zaytzef rule we know elimination predominantly occurs from more substituted group. so we get A is major product.



Usually elimination occurs from trans periplanar position. But in this case, the reaction proceeds through SNi reaction and gives equatorial alcohol as major product. Because equatorial alcohol comes from stable conformer.